Zhankun Luo

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Research Interests

My research focuses on developing robust and efficient ML systems by bridging theoretical analysis with practical application: I design algorithms for large-scale optimization (constrained optimization) and statistical learning (variance-reduced methods, Expectation-Maximization), and deploy ML/CV solutions for manufacturing, dietary assessment, and structured light vision.

Education

Purdue University West Lafayette — West Lafayette, IN
Ph.D. in Electrical and Computer Engineering — *Aug 2021 – Present*

Purdue University Northwest — Hammond, IN
Master in Electrical and Computer Engineering — *Aug 2019 – May 2021*

Beijing Institute of Technology — Beijing, China
Bachelor in Telecommunication Engineering — *Sept 2015 – Jun 2019*

Research Experience

Research Assistant (PI: Prof. Abolfazl Hashemi) — West Lafayette, IN
Machine Intelligence & Networked Data Science (MINDS) Group, Purdue University West Lafayette — *Aug 2025 – Present*

- Formulated efficient, theoretically grounded algorithms as part of project "Learning through the Air: Cross-Layer UAV Orchestration for Online Federated Optimization", enabling practical deployment of learning-based systems under resource constraints

Research Assistant (PI: Prof. Edward J. Delp and Prof. Fengqing M. Zhu) — West Lafayette, IN
Video and Image Processing Laboratory (VIPER), Purdue University West Lafayette — *Aug 2021 – May 2022*

- Investigated methods for fine-grained food classification for "Technology Assisted Dietary Assessment (TADA)" and incorporated synthetic UAV image generation to improve panicle detection for "Image Based Plant Phenotyping: The PhenoSorg Project"

Research Assistant (PI: Prof. Chenn Zhou, Prof. Bin Chen and Prof. Lizhe Tan) — Hammond, IN
Center for Innovation through Visualization and Simulation (CIVS), Purdue University Northwest — *Feb 2020 – May 2021*

- Developed an ML solution for casting temperature prediction ("Smart Ladle: Al-Based Tool for Optimizing Casting Temperature") and proposed a robust framework for object height measurement using "Structured Light Vision Systems"

Project

Convergence Analysis and Algorithmic Design for Constrained Optimization Methods — West Lafayette, IN
Machine Intelligence & Networked Data Science (MINDS) Group, Purdue University West Lafayette — *Aug 2025 – Present*

- Established convergence guarantees for switching methods in Softmax-weighted minimax optimization with constraints
- Formulated a unified framework for variance-reduced constrained optimization**(Skill: Optimization Theory, Stochastic Analysis)**

Characterizing Evolution in EM Estimates for Overspecified MLR — West Lafayette, IN
Machine Intelligence & Networked Data Science (MINDS) Group, Purdue University West Lafayette — *Nov 2023 – Aug 2025*

- Developed a theoretical understanding of EM under model misspecification for 2MLR, establishing linear convergence with an unbalanced initial guess and sublinear convergence with a balanced initial guess
- Bridged population and finite-sample results to derive iteration complexity bounds and extended the analysis to the low SNR regime
- Demonstrated distinct statistical accuracy bounds at the finite-sample level for mixtures with sufficiently unbalanced versus sufficiently balanced fixed mixing weights **(Skills: Non-Asymptotic Statistical Analysis, Dynamical Systems, Mathematica)**

Structural Properties, Cycloid Trajectories and Non-Asymptotic Guarantees of EM for MLR — West Lafayette, IN
Machine Intelligence & Networked Data Science (MINDS) Group, Purdue University West Lafayette — *May 2023 – Oct 2025*

- Provided explicit closed-form expressions of EM updates in two-component Mixed Linear Regression (2MLR) under all SNR regimes
- Characterized the behavior of EM iterations and notably showed that all the iterations lie on a certain cycloid
- Exhibited theoretical estimate for the exponent of super-linear convergence and enhanced the statistical error bounds**(Skill: Python)**

Technology Assisted Dietary Assessment (TADA) — West Lafayette, IN
Video and Image Processing Laboratory (VIPER), Purdue University West Lafayette — *Aug 2021 – May 2022*

- Investigated reliable and effective methods for Fine-Grained Visual Classification (FGVC)
- Re-implemented a hierarchy-based embedding method for encoding of categories to decrease average hierarchical distance at top 1 by 3%, and that at top 5 by 10% on our VIPER-FoodNet dataset with 82 food categories, 15 thousand images
- Worked on improving hierarchical methods by incorporating the nutrient and visual information of food **(Skills: Python, PyTorch)**

Image Based Plant Phenotyping: The PhenoSorg Project — West Lafayette, IN
Video and Image Processing Laboratory (VIPER), Purdue University West Lafayette — *Aug 2021 – May 2022*

- Generated 1 thousand synthetic high-resolution UAV RGB images with panicle labels by using image-to-image translation GANs with a ground truth dataset of 400 real UAV RGB images; improved mean average precision from 0.5 to 0.95 for panicle detection task from 72% to 79%, and reduced Mean Absolute Percent Error (MAPE) for panicle counting task from 11.6% to 7.2%
- Worked on creating labels for panicles in PhenoRover RGB images to test our approach on data

Smart Ladle: AI-Based Tool for Optimizing Casting Temperature Hammond, IN

Center for Innovation through Visualization and Simulation (CIVS), Purdue University Northwest *Feb 2020 – May 2021*

- Developed a machine learning application using DNN, lightGBM to provide steel casting temperature predictions
- Reduced Root Mean Square Error (RMSE) of predicted casting temperature to 3 degrees Fahrenheit
- Collaborated application with SQL database and GUI using Unity (C#) to display predictions and parameters
- Tested and deployed this tool at Steel Dynamics Inc (SDI) Butler Division, awarded AIST 2022 Hunt-Kelly Outstanding Paper Award – third place (AIME) and AIST 2021 Digitalization Applications Technology Best Paper Award **(Skills: Python, SQL, Bash, C#, Unity)**

Structured Light Vision Systems Hammond, IN

Department of Electrical and Computer Engineering, Purdue University Northwest *Aug 2019 – May 2021*

- Proposed a multi-level RANSAC method to tackle intersections of multiple lasers which decreased MAPE by 20%
- Developed a framework for height measurement using a structured light system with multiple cameras and multiple laser emitters whose Mean Absolute Percent Error (MAPE) reached 4% **(Skills: Multiple View Geometry, Camera Calibration, MATLAB, C++)**

Automated Fetal Brain Segmentation Using Deep Convolutional Neural Network Hammond, IN

Department of Electrical and Computer Engineering, Purdue University Northwest *Sept 2018 – Jan 2019*

- Designed a U-Net based method with generative adversarial loss for automatic fetal brain segmentation, evaluated the performance with FPN, U-Net architectures and GAN loss, focal loss
- Labeled 6.7 thousand fetal brain slice images from coronal, transverse, and sagittal MRI scans from 30 to 33 weeks to be used for autonomous fetus brain registration **(Skills: Data Processing, Python, PyTorch, MATLAB, Git)**

Parking Occupancy Detection Using Computer Vision Hammond, IN

Department of Electrical and Computer Engineering, Purdue University Northwest *Aug 2018 – May 2019*

- Developed a software in Python (PyTorch) based on CNN to identify and count occupied parking spaces, achieving 90% accuracy and 96% precision on videos of parking spaces
- Incorporated the mutual information method for image registration and the affine transformation to eliminate the impact caused by camera shake and enhance the robustness of detection **(Skills: Image Processing, Python, PyTorch, Git)**

Technical Skills

Key Courses: Optimization for Deep Learning, Statistical Machine Learning, Probabilistic Machine Learning, Model-Based Image and Signal Processing, Digital Image and Signal Processing, Estimation Theory, Reinforcement Learning, Digital Control Systems Analysis, Pattern Recognition and Decision-Making Processes, Lumped System Theory, Introduction to Mathematical Statistics, Real Analysis
Programming: Python, C/C++, MATLAB, Bash, SQL, Java, C#, Mathematica, HTML, LaTeX
Frameworks: DL (PyTorch, TensorFlow), Data Processing (NumPy, Scipy, scikit-learn, pandas), Computer Vision (OpenCV, eigen3, Sophus)
Platforms: GNU/Linux, Windows, MacOS
Tools: Git, gdb, CMake, Colab, Qt, Tableau, Visio, Inkscape, Unity, Nsight, PSpice, AutoCAD, Autodesk Inventor
IDEs: VS Code, PyCharm, Visual Studio, IntelliJ, Eclipse, KDevelop, Android Studio

Publications & Working Papers

[1] **Z. Luo**, A. Hashemi. *"Unifying Score Matching and Expectation-Maximization via KL Divergence in Mixed Linear Regression,"* Manuscript in preparation.

[2] **Z. Luo**, D. Lee, A. Hashemi, and A. Makur. *"Strong Antithetic Variates: Theory and Applications,"* Manuscript in preparation.

[3] A. Hashemi, **Z. Luo**, S. B. Moon, M. B. Sahin, and A. Upadhyay. *"High-Probability Dynamics of Variance-Reduced Estimators: A Unified Analysis,"* Manuscript in preparation.

[4] **Z. Luo**, A. Upadhyay, S. B. Moon, and A. Hashemi. *"First-Order Softmax Weighted Switching Gradient Method for Distributed Stochastic Minimax Optimization with Stochastic Constraints,"* Preprint (under review), Submitted to *Conference on Uncertainty in Artificial Intelligence (UAI).*

[5] **Z. Luo** and A. Hashemi. *"Structural Properties, Cycloid Trajectories and Non-Asymptotic Guarantees of EM Algorithm for Mixed Linear Regression,"* Preprint (under review), Submitted to *IEEE Transactions on Information Theory (IT).*

[6] **Z. Luo** and A. Hashemi. *"Characterizing Evolution in Expectation-Maximization Estimates for Overspecified Mixed Linear Regression,"* Proceedings of *Transactions on Machine Learning Research (TMLR)*, Jan 2026.

[7] **Z. Luo** and A. Hashemi. *"Unveiling the Cycloid Trajectory of EM Iterations in Mixed Linear Regression,"* Proceedings of *International Conference on Machine Learning (ICML)*, July 2024.

[8] E. Cai, **Z. Luo**, S. Baireddy, J. Guo, C. Yang, and E. J. Delp. *"High-Resolution UAV Image Generation for Sorghum Panicle Detection,"* Proceedings of *the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, June 2022.

[9] N. J. Walla, **Z. Luo**, B. Chen, Y. Krotov, and C. Q. Zhou. *"Smart Ladle: AI-Based Tool for Optimizing Caster Temperature,"* Proceedings of *the Iron & Steel Technology Conference (AISTech)*, May 2021.

[10] **Z. Luo**. *"Structured Light Vision Systems Using a Robust Laser Stripe Segmentation Method,"* M.S. thesis. Purdue University, May 2021.
[11] **Z. Luo**, Y. Zhang, and L. Tan. *"Multi-Level Random Sample Consensus Method for Improving Structured Light Vision Systems,"* Proceedings of *IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*, Oct 2020.
[12] Y. Zhang, **Z. Luo**, J. Hou, L. Tan, and X. Guo. *"Computer Vision Techniques for Improving Structured Light Vision Systems,"* Proceedings of *IEEE International Conference on Electro Information Technology (EIT)*, Aug 2020.

Positions of Responsibility

Graduate Student Mentor - English Training in Engineering (ETIE)
Developed project frameworks in machine learning and computer vision, and mentored undergraduate students working on related topics, including sequential prediction and 3D reconstruction

Coordinator and Host – Science Communication Contest
Contacted and coordinated with participants and judges, and organized a science lecture competition for 3,600+ students

Team Leader – Math Modeling Contests
Led a team in national and international competitions, developing and implementing algorithms such as SVM, K-means, Markov chains, LDA, PCA, time series analysis, and multivariable regression; awarded the Second Prize in CUMCM and an Honorable Mention in ICM

Teaching Experience

Teaching Assistant (Instructors: Prof. Abolfazl Hashemi and Prof. Joseph Makin) West Lafayette, IN
ECE Department, Purdue University West Lafayette *May 2022 – Aug 2025*

- Mentored students on the foundational theory and analysis of stochastic gradient-based algorithms, advanced training methods, and distributed learning for the course "ECE69500 Optimization for Deep Learning"
- Held office hours and help sessions, graded homework assignments, wrote exam problems and homework solutions, and composed helpful materials for "ECE20001 Electrical Engineering Fundamentals I" and 'ECE20002 Electrical Engineering Fundamentals II"

Professional Services

Journal and Conference Reviewers: IEEE Transactions on Signal Processing (TSP),
International Conference on Learning Representations (ICLR), Conference on Uncertainty in Artificial Intelligence (UAI),
International Conference on Artificial Intelligence and Statistics (AISTATS), IEEE International Symposium on Information Theory (ISIT),
IEEE International Conference on Multimedia and Expo (ICME), Annual Conference on Vision and Intelligent Systems (CVIS)
Memberships: IEEE Young Professionals, IEEE Signal Processing Society, IEEE Computer Society

Awards and Honors

Conference Travel Grant for NeurIPS 2025	2025
AIST 2022 Hunt-Kelly Outstanding Paper Award – third place (AIME)	*2022*
AIST 2021 Digitalization Applications Technology Best Paper Award	*2021*
Innovative Practice Outstanding Student (5%)	*2019*
National third prize of Energy Saving and Emission Reduction Social Practice and Technology Competition	*2017*
Honorable Mentioned prize of the Interdisciplinary Contest in Modeling (ICM)	*2017*
Second prize of Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)	*2016*